

PixelForge: Responsive Gaming Hub

Team: PixelForge

Group: SE-2501

Team members: Marat Nurdaulet, Kadyr Tlektes, Suinalin Azamat

Objective

Extend the existing Assignment 1 website using Flexbox for one-dimensional alignment and CSS Grid for page structure and a gallery. Preserve the six pages, dark gaming theme, original content and local images while making the site responsive without a CSS framework.

Live website: <https://nurda07.github.io/Web-assignment/>

Repository: github.com/NURDA07/Web-assignment

Scope and page ownership

Marat Nurdaulet: Home and About.

Kadyr Tlektes: Top Games and Guides.

Suinalin Azamat: Team and Contact.

The shared stylesheet applies the new header, sidebar, Grid structure and footer to every page.

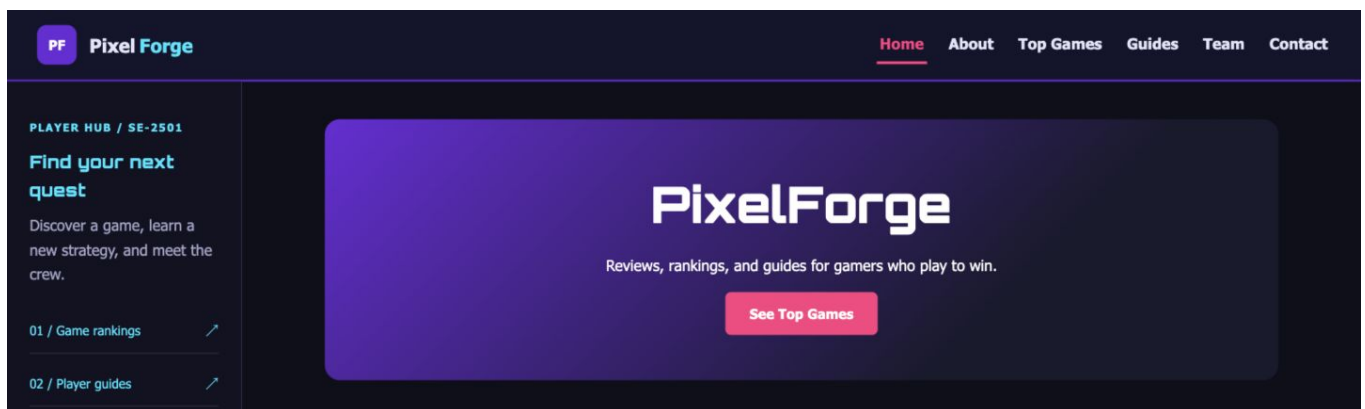


Figure 1. The existing PixelForge identity with the new shared navigation and sidebar.

Deliverables

Six updated HTML pages, a shared CSS stylesheet, local images, a small contact-draft script, an automatic GitHub Pages deployment workflow, this report and a ZIP of the project.

Task 1: Flexbox Navigation

Goal: Place the logo on the left and navigation links on the right, with consistent spacing and vertical alignment.

Steps

1. Added a linked PixelForge brand and PF badge inside each existing header.
2. Applied `display: flex` to `#main-header`.
3. Used `justify-content: space-between` to separate logo and links.
4. Used `align-items: center` and `gap` for alignment and spacing.
5. Kept the navigation list as a nested Flexbox container with wrapping. On small screens, the header stacks and links wrap.
6. Added an active-page indicator, navigation labels and a keyboard skip link.

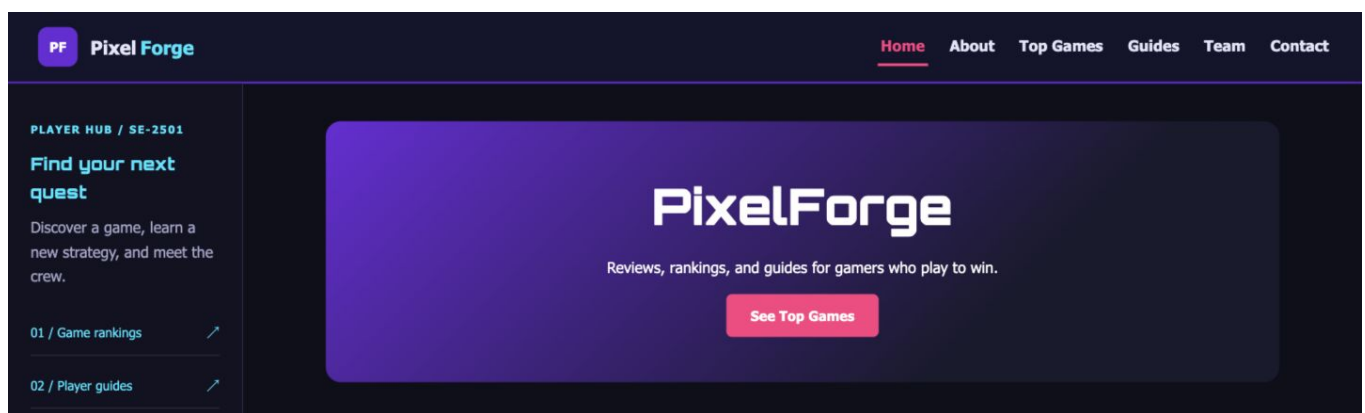


Figure 2. Desktop navigation: logo on the left, links on the right, vertically centered.

Relevant CSS

```
#main-header { display: flex;  
align-items: center; justify-content: space-between;  
flex-wrap: wrap; gap: 1rem 2rem; }
```

Result

All six pages use the same navigation layout. The current page is marked visually and with `aria-current`. Links remain visible at 390px without horizontal page overflow.

Task 2: Equal-height Flex Cards

Goal: Display at least three cards in a row. Each card contains an image, title, text and a button, with equal heights and a hover effect.

Steps

1. Converted the existing card containers to `.card-row` on Home, Guides and Team.
2. Added missing images and working links styled as buttons.
3. Applied Flexbox, wrapping, a consistent gap and `align-items: stretch` to each row.
4. Made each card a vertical flex container. `margin-top: auto` aligns the buttons at the bottom.
5. Added a subtle lift and shadow for hover and keyboard focus.
6. Switched to two columns at tablet widths and one column on mobile.

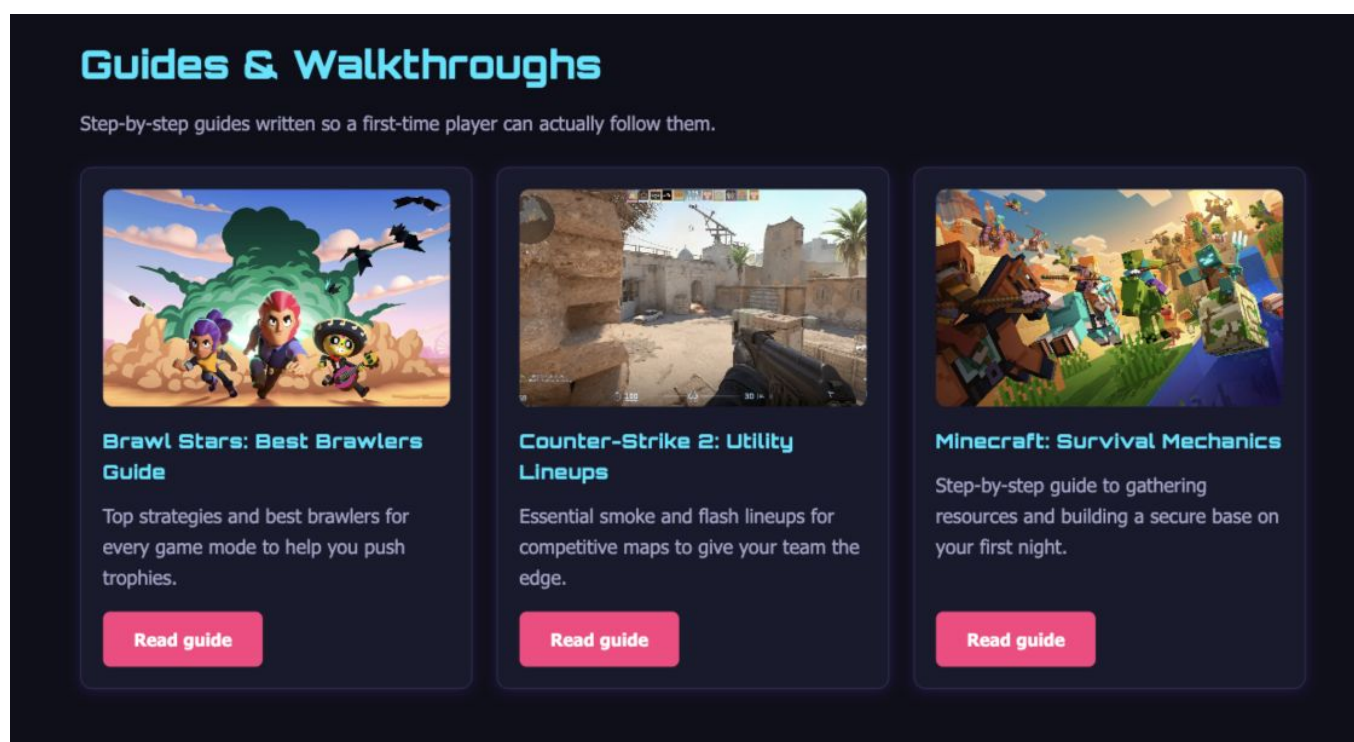


Figure 3. Three guide cards with equal heights and aligned buttons. Buttons lead to guide sections.

Validation

At 1440px, browser measurements were 381px for all three Home cards, 432px for all three Guides cards and 367px for all three Team cards. Equal height applies within each flex row. Images retain a consistent aspect ratio without stretching.

Task 3: Named Grid Areas

Goal: Arrange a full-width header, left sidebar, right main content and full-width footer.

1. Applied display: grid to the body on every page.
2. Defined two columns with a bounded sidebar and flexible main column.
3. Defined three rows: auto 1fr auto.
4. Named areas as "header header" "sidebar main" "footer footer".
5. Assigned each semantic element its matching grid-area.
6. Below 700px, replaced the layout with one column: header, sidebar, main, footer.

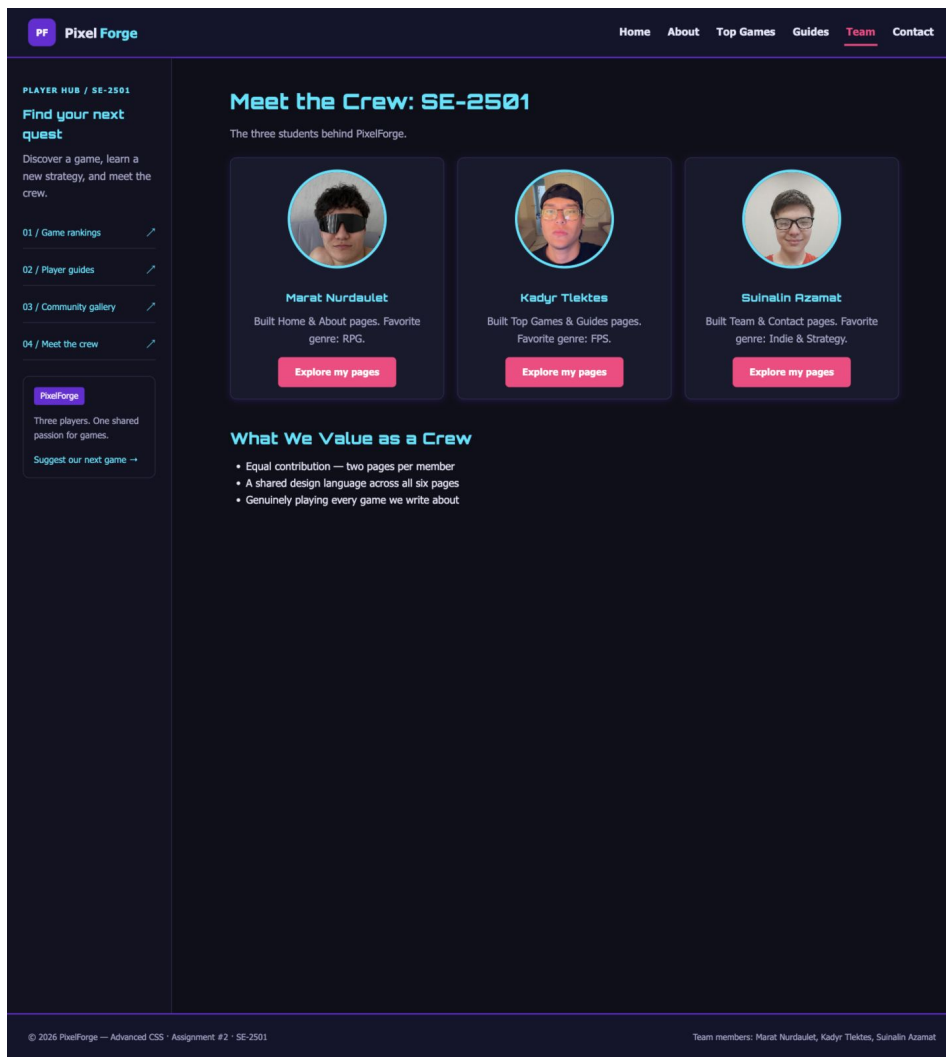


Figure 4. Team page showing all four named areas, including the full-width footer.

The main column uses `minmax(0, 1fr)` and `min-width: 0` to avoid overflow from long content. The sidebar provides useful links to rankings, guides, gallery and team pages.

Task 4: Image Gallery

Goal: Build a gallery with at least nine images, equal Grid tracks, consistent gaps and caption overlays.

1. Reused nine distinct local images already in the repository: three games, a game collage, two PixelForge images and three team portraits.
2. Added semantic figures and captions in the Top Games page.
3. Used `repeat(3, minmax(0, 1fr))`, equal 210px rows and a 1rem gap.
4. Applied object-fit: cover to preserve image proportions.
5. Revealed captions on hover or keyboard focus. Touch devices show them by default.
6. Reduced the gallery to two columns on tablets and one below 420px.

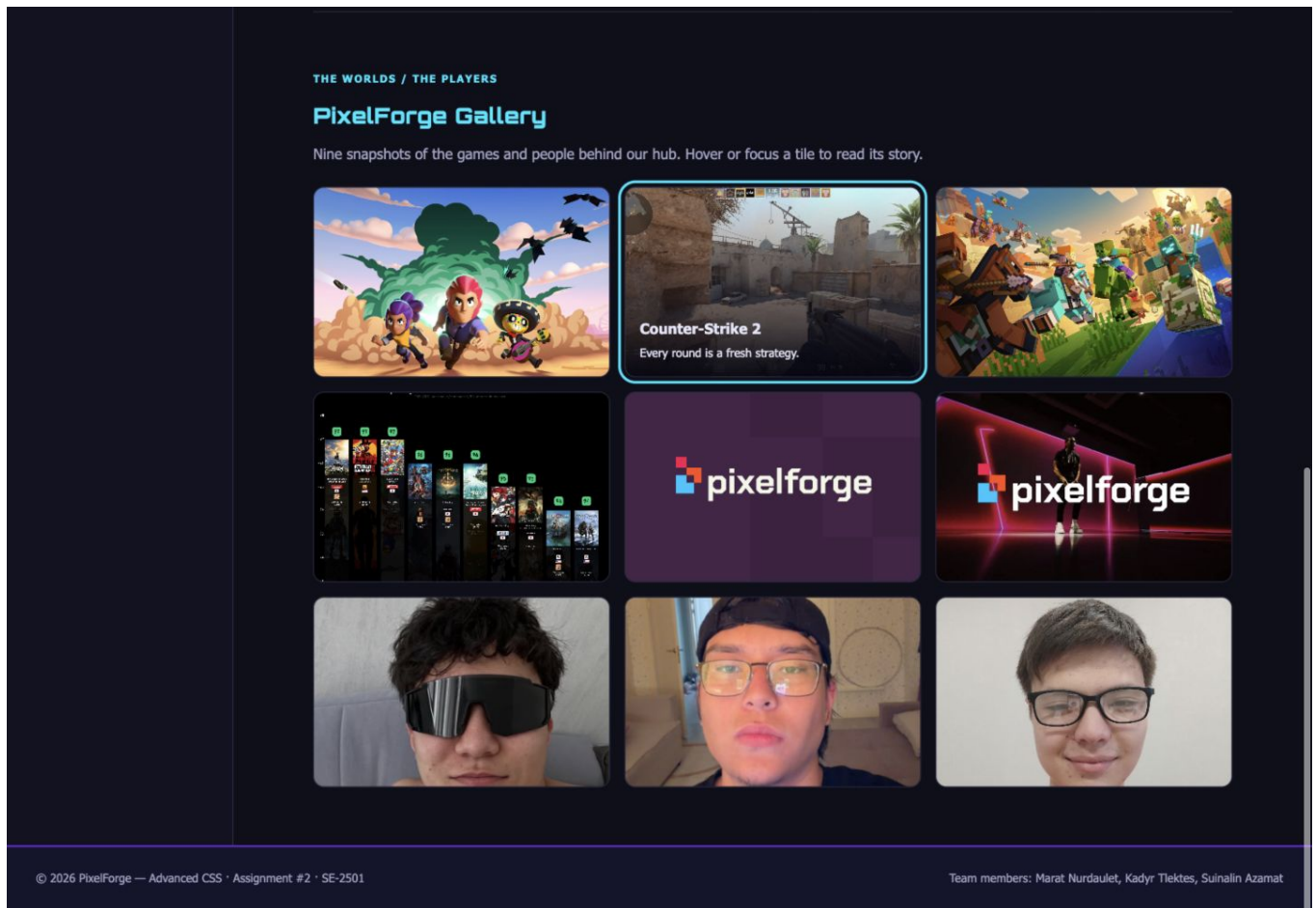


Figure 5. All nine tiles. Keyboard focus on Counter-Strike 2 reveals its caption and visible focus outline.

The original project images remain local, so the gallery does not depend on external image hosts. The focus behavior was checked in the browser: the focused caption reached opacity 1.

Part 3: Responsive Integration

The six pages share the same colors, spacing, navigation and Grid structure. Flexbox handles navigation and cards; Grid handles the page and gallery. No external layout framework is used.

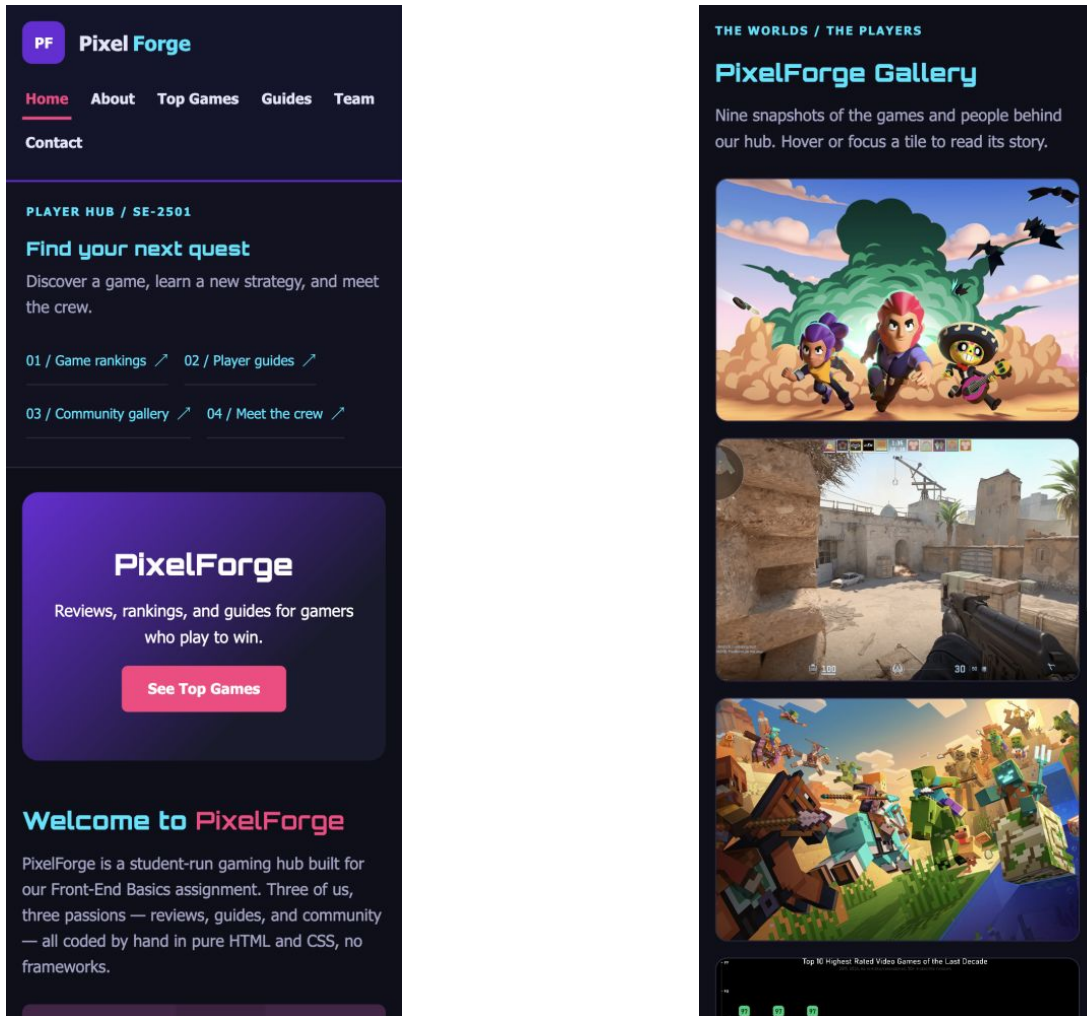


Figure 6. Home navigation and single-column gallery at a 390px viewport.

Browser and file checks

All six pages were inspected at 1440px and 390px: no horizontal page overflow. Desktop images loaded correctly. All local links, image paths and fragment targets resolve. The ranking table scrolls inside its own container on narrow screens. The contact form validates inputs and prepares a local draft, with an explicit message that nothing has been sent.

Accessibility improvements include descriptive image text, visible focus outlines, a skip link, labelled navigation, keyboard-accessible gallery captions and reduced-motion support.

Existing Pages Preserved

About retains the project story and values. Contact retains the contact information and form fields, uses an existing local image, and prepares a draft instead of making a failing static-site POST request.

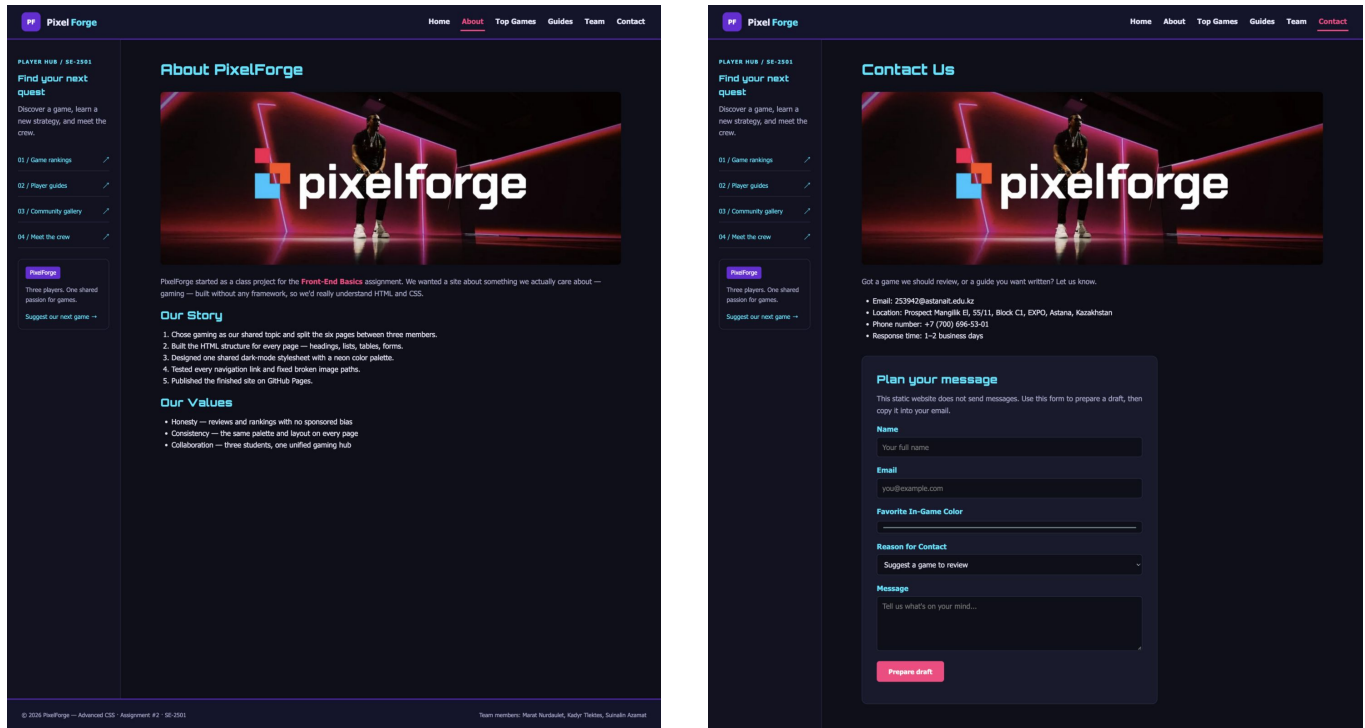


Figure 7. About and Contact after the shared layout update.

What changed on every page

The existing navigation gained a logo and Flexbox header. A shared sidebar was added. The main content and footer were assigned Grid areas. Footer text now identifies Assignment 2 and SE-2501. The original author comments and page content remain in the project.

Engagement

Home cards lead to rankings, guides and the team. Guide cards lead to readable step-by-step sections. Team cards link to the members' pages. Gallery overlays describe the games and people behind the project.

Part 4: Publication and Reflection

Deployment steps

1. Committed the website, README and deployment workflow in the existing repository.
2. Pushed commit 597ac3d to the main branch.
3. Enabled GitHub Pages with the GitHub Actions build type.
4. The workflow checked out the repository and uploaded the pixelforge directory as the site artifact.
5. GitHub Pages deployed the artifact successfully. The live homepage was opened and verified.

Live URL: <https://nurda07.github.io/Web-assignment/>

Successful deployment: [GitHub Actions run 35918010090](#)

Reflection

Flexbox is well suited to aligning a row of links and distributing space inside cards. Grid makes the relationship between header, sidebar, main content and footer explicit. Combining them keeps layout code readable and makes responsive changes predictable. Reusing the original pages and assets preserves the identity of Assignment 1 while meeting the new layout requirements.

Preparation for the practical defense

Explain the main and cross axes of Flexbox, justify-content versus align-items, the role of gap, and why margin-top: auto aligns card buttons. Demonstrate the named Grid areas, repeat() and minmax(), then resize the browser to explain the media queries. Use Tab to reveal a gallery caption and open a guide through a card button.

Submission checklist

Submit this PDF, PixelForge_Assignment_2_Project.zip and the live URL. Each team member must submit through the LMS by the course deadline and attend the practical defense. The archive includes the complete project, README, deployment workflow and a copy of the report. No LMS submission has been performed automatically.